

RMD-Q Yellowpaper
— Formal Specification v0.1
(DRAFT: research prototype, not a standard)

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Abstract

Formal specification of the RMD-Q KEM (fully specified) and signature v1 (toy status): notation, the assumption with its planted distribution, frozen parameter sets, exact algorithms, wire formats, KAT inventory, number-theoretic transform, and the claims/gap table. Companion to the whitepaper (vision) and the ePrint report.

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1 Notation

$R_q = \mathbb{Z}_q[X]/(X^n + 1)$, n power of two, q prime; production $(n, q) = (256, 3329)$. $S(\eta, w)$: at most w nonzero coefficients per polynomial, values uniform over $[-\eta, \eta]$ then sparsified (secrets only). CBD(η): centered binomial, fixed-shape (errors, encryption randomness). Symmetric layer: SHAKE128/256 (FIPS 202) with fixed domain separation (0x00 matrix, 0x01 FO coins/shared-secret, 0x02 signatures, **reject** implicit rejection). $\|\cdot\|_\infty$ centered; $\text{wt}(\cdot)$ Hamming weight. **encode**: 32 message bytes to one polynomial ($q/2$ vs 0 per bit); correct iff per-coefficient noise $< q/4$.

2 Assumption RMD-Q

Definition 1 (Search-RMD-Q). *Parameters* $(n, q, k, \eta, w, \eta_e, t, m)$, $N = k \cdot n$. **Gen** samples $A \leftarrow U(R_q^{k \times k})$, $s \leftarrow S(\eta, w)^k$, $e \leftarrow \text{noise}(\eta_e)$, $P \leftarrow \text{MQ}(t, m)$ conditioned on $P(s) = 0$, sets $b = As + e$. Given (A, b, P) , find s' with $\text{wt} \leq kw$, $\|s'\|_\infty \leq \eta$, $\|As' - b\|_\infty \leq B_e$, $P(s') = 0$ exactly. Decision variant distinguishes $(A, As + e, P)$ from uniform.

P is **planted** (sample s , construct P around it: $m-1$ free terms plus one solved fix monomial on support(s), uniform slot, resample-until-nonzero with pure-random escape) and **homogeneous-quadratic** (load-bearing for the signature opening). Measured fix-coefficient $\chi^2 = 17.4$ (null mean 15, $n = 209$).

3 Parameters (frozen v0.1)

set	n	q	k	η	w/poly	η_e	t	m	wire
TOY-16	16	17	1	1	4	1	2	4	raw
TOY-32	32	97	1	1	6	1	3	6	raw
PROD-256	256	3329	2	2	128	2	8	12	12-bit
PROD-384	256	3329	3	2	160	2	10	12	estimates only

4 KEM

KeyGen(*seeds*): (1) expand A via SHAKE128; (2) sample sparse s , plant P around s , sample CBD error e ; (3) $b = As + e$; output $pk = (\text{seed}_A, \text{seed}_P, b)$, $sk = (s, pkhash, \sigma)$. **Encaps**(pk): (1) $m \leftarrow \{0, 1\}^{256}$; (2) *coins* from SHAKE256 over $(m, pkhash)$, sample dense CBD (r, e_1, e_2) ; (3) $u = A^T r + e_1$, $v = b^T r + e_2 + \text{encode}(m)$; (4) K derived alongside ct . **Decaps**(sk, ct): decode $m' = v - s^T u$, re-encrypt, byte-compare; match returns K , else $K = \text{SHAKE256}(\text{reject} \parallel \sigma \parallel ct)$ (implicit rejection). Wire v0.1: 12-bit lossless packing (NTT-friendly $q < 4096$); compressed $(d_u, d_v) = (10, 5)$ analyzed ($2^{-268.6}$) with Python FO path proven, C codec proven. KATs: `kat_toy16`, `kat_fo16`, `(cbd)`, `kat_kem256`, `(cbd)`.

5 Signature v1 (toy)

Fiat-Shamir with dual opening from one mask y : $z_{\text{lat}} = y + c_{\text{poly}}s$ (lattice check) and $z_{\text{mq}} = y + c_{\text{scalar}}s$ with $P(z_{\text{mq}}) = P(y) + ch$ (MQ check). Prod (τ, γ_1, β) , wire sizes, proofs: unspecified (toy).

6 NTT (valid (256, 3329) only)

7-layer DIT, $k=1$ skipped, odd-power pair moduli $z_i = w^{2^{\text{rev}_7(i)+1}}$, Barrett $\mu=20158$ exhaustively proven over all 11M products, inverse table, constant-time twins (0 sign-jumps, automated smoke check).

7 Claims, non-claims, gaps

Claimed: KAT-level correctness, FO mechanics, toy breakability as predicted, measured hygiene, relative model estimates. NOT claimed: NIST categories, reductions/proofs, assumption certification, side-channel resistance beyond timing smoke, signature security, deployment readiness. Full gap table: `spec/07-evidence.md`.